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Paper Code : PCC-CS-301

PUID : 03443 (To be mentioned in the main answer script)

DATA STRUCTURE & ALGORITHMS

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

i) The worst case complexity of binary search for a list having n elements is

a) $\log 2^n$ b) $n \log 2^n$

c) n d) n^2.

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ii) Which among the following are applications of queues ?

a) Queues keep track of events waiting to be handled, like multiple button clicks

b) Queues are used for evaluation of arithmetic expressions

c) Queues are used in parsing

d) None of these.

iii) A full binary tree with n non-leaf nodes contains

a) $\log 2n$ nodes b) $n+1$ nodes

c) $2n$ nodes d) $2n+1$ nodes.

iv) Consider that n elements are to be stored. What is the worst case time complexity of Bubble sort ?

a) $O(1)$ b) $O(\log 2n)$

c) $O(n)$ d) $O(n\text{^}2)$.

v) The pre-requisite condition of binary search is

a) Unsorted array

b) Ascending order array

c) Descending order array

d) Sorted array.

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vi) The following sequence of operations are performed on a stack : push (1), push (2) , pop, push (1), push (2), pop, pop, pop, push (1) pop. The sequence of popped out values are

a) 2, 2, 1, 1, 1 b) 2, 2, 1, 2, 2

c) 2, 1, 2, 2, 1 d) 2, 1, 2, 2, 2.

vii) The postfix equivalent of the prefix expression $*+ab-cd$ is

a) $ab+cd-*$ b) $abcd+-*$

c) $ab+cd*-$ d) $ab+-cd*$.

viii) Linked list is a

a) Linear data structure

b) Dynamic data structure

c) Self referential data structure

d) all of these.

ix) One limitation of linked list is

a) it requires huge memory space

b) it requires contiguous memory space

c) nodes can only be accessed sequentially

d) nodes can only be accessed randomly.

x) What is the worst case performance of Selection sort algorithm ?

a) $O(\log n)$ b) $O(n*n)$

c) $O(n)$ d) $O(n \log n)$.

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xi) A sort which relatively passes through a list to exchange the first element with any element less than it and then repeats with a new first element is called

a) insertion sort b) selection sort

c) heap sort d) quick sort.

xii) Recursion uses more memory space than iteration because

a) it uses stack instead of queue

b) every recursive call has to be stored

c) both (a) and (b)

d) none of these.

GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. Write an algorithm for Binary Search with non-recursive function call.

3. Write the algorithms for PUSH and POP for linked representation of stack. $2\frac{1}{2} + 2\frac{1}{2}$

4. Explain with suitable example the collision resolution scheme using linear probing with open addressing.

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5. Write an algorithm to delete a node having value NUM from a Doubly linked list.

6. Sort the following data using bubble sort algorithm (show the intermediate outputs) :

12, 32, 23, 15, 34, 11, 25.

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. a) Explain Dijkstra's algorithm for finding the shortest distance between two given nodes. Consider a graph of your choice having at least five nodes and specific weight assigned to each edge.

b) What is priority queue ?

c) How can the polynomial $4x^6 + 7x^3 + 2$ be represented using a linked list and an array ? Also explain why arrays should not be used in this regard.

d) What is the necessity of having circular queue ?

$7 + 2 + 4 + 2$

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8. a) How does AVL tree differ from binary search tree ?

b) Explain different rotation mechanisms in AVL tree.

c) Build an AVL tree with nodes inserted in the following order :

8, 12, 9, 11, 7, 6, 62, 15, 3

Clearly mention different rotations used and balance factor of each node.

d) Show that for a non-empty binary tree, if n_0 is the number of leaf nodes and n_2 is the number of nodes of degree 2 then $n_0 = n_2 + 1$. 3 + 3 + 4 + 5

9. a) Create a binary search tree using the following data elements : http://www.makaut.com

54, 60, 69, 32, 46, 25, 81, 29, 75, 78

b) Explain with example the process of deleting a node from a binary search tree.

c) Write down the algorithm for deleting a node from a binary search tree considering all possible cases of number of children. 4 + 5 + 6

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10. a) Explain linear and non-linear data structure with example.

b) Explain different types of data structure operations.

c) What is meant by algorithmic complexity ?

d) What is time-space trade-off ? 4 + 4 + 4 + 3

11. a) Explain Selection Sort algorithm.

b) Write an algorithm for Selection Sort.

c) Compare Selection Sort with Insertion Sort.

d) Sort the following data using selection sort algorithm (show the intermediate outputs) :

23, 12, 21, 15, 14, 50, 13. 3 + 4 + 4 + 4

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